

ML-based air quality monitoring and health risk prediction for construction worker safety.

Construction × Worker safety monitoring × Machine learning · OS112 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
54 is the world moving	48 can we play, can we win	MEDIUM 0 named in the evidence	€176m observed evidence · per year	NOW budgeted procurement within 1...	L1 12 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 1.0 published reference(s) in Construction, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

We can help construction firms protect workers from air quality hazards using machine learning to predict health risks in real time. This opportunity exists now because ML models have been proven to monitor and predict air quality at construction sites, and the industry is under pressure to improve safety outcomes.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€394m €29.2m – €1.6bn	€176m €13.1m – €703m	€2.1m €157k – €8.4m	observed
Observed: tendered contract value, annualised	€123m €123m – €471m	€123m €123m – €471m	€1.5m €1.5m – €5.6m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Construction (EU27_2020)	250,045 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Machine learning	1.8 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_ain2 · E_AI_TML	2025
Annual value of one engagement	€952k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	22,551 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 2 of 2 declared geographies are outside the European reference data (AE, CN); the estimate covers the European footprint only.

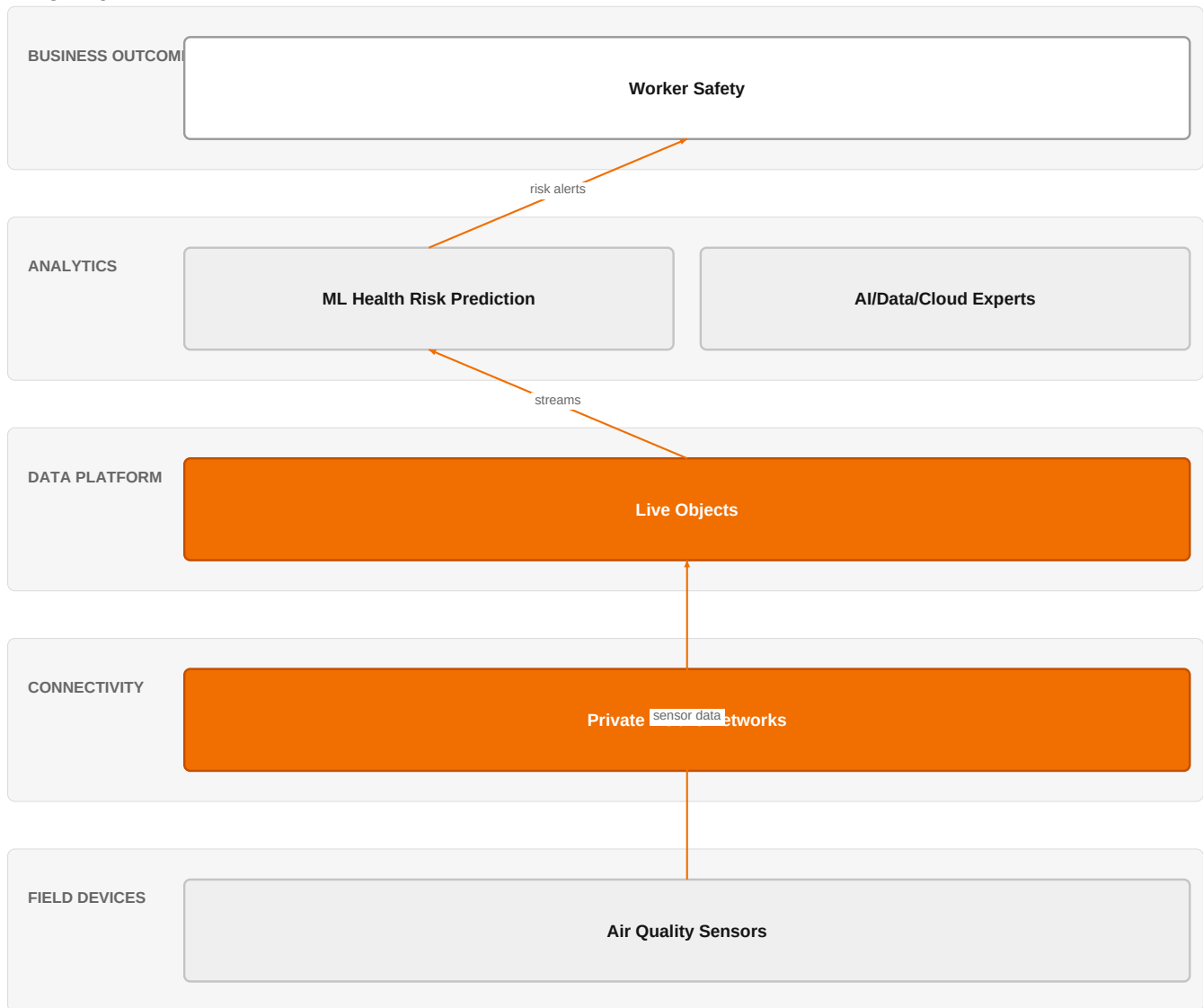
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€123m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 24 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 2 of 2 declared geographies are outside the European reference data (AE, CN); the estimate covers the European footprint only.

The solution, in one picture

Air Quality Health Risk Prediction for Construction



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This solution collects air quality data from construction sites, processes it in the cloud, and delivers predictive health risk alerts to improve worker safety.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Research demonstrates that air quality at construction sites significantly impacts worker health and safety, and machine learning models can effectively monitor and predict these risks (SIG-6ADB457BBC36). Additionally, interpretable machine learning is emerging as a tool to understand and improve safety behaviors, overcoming limitations of traditional methods (SIG-9528B2CFF302). The construction industry remains among the most hazardous sectors, and AI is being systematically reviewed for safety management (SIG-3E11F4C71251).

Sources: SIG-6ADB457BBC36 openalex.org 2025-11-19; SIG-9528B2CFF302 openalex.org 2026-01-01; SIG-3E11F4C71251 openalex.org 2025-11-13

Who buys, and why now

The buyer is typically a construction firm's safety director or operations manager, who feels the pain of managing worker safety on sites with variable air quality. The trigger is often a near-miss incident, a regulatory inspection, or a new project in an area with known pollution concerns. They need to reduce health risks and avoid costly stoppages or liability.

Why it is hot now — every claim bound to a dated source

- Air quality at construction sites significantly impacts workers' health and safety, and machine learning models can monitor and predict air quality and health risks. [SIG-6ADB457BBC36]
- Unsafe behaviors among construction workers remain a leading cause of accidents, and interpretable machine learning can explore determinants of safety behavior, overcoming limitations of traditional methods. [SIG-9528B2CFF302]
- Machine learning is used to predict air quality and health risks at construction sites. [SIG-6ADB457BBC36]
- Machine learning models can monitor and predict air quality and associated health risks for construction workers. [SIG-6ADB457BBC36]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its Live Objects IoT platform to collect air quality sensor data from construction sites, and its AI/Data/Cloud experts to build and deploy machine learning models that predict health risks. Private mobile networks could ensure reliable connectivity on site. The engagement would start with a pilot on one site, then scale across the customer's portfolio.

Why Orange can win

Orange can win by combining its IoT platform (Live Objects) with its AI/Data/Cloud expertise and private mobile networks, offering an integrated solution that hyperscalers may not provide end-to-end. However, the assets are not uniquely differentiated; competitors like AWS and Google Cloud also offer ML platforms, and Atos/Eviden has systems integration capabilities.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer provides the technology in a matching domain	unconfirmed
Private mobile networks	offer	L1	offer addresses the use case but not this technology	unconfirmed
AWS	partner	L2	partner provides the required technology	unconfirmed
Google Cloud	partner	L2	partner provides the required technology	unconfirmed
NICE	partner	L2	partner provides the required technology	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 7.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells Machine learning; competes in OX: Smart Industries — also an Orange partner	structural	—
Google Cloud	Hyperscaler	sells Machine learning — also an Orange partner	structural	—
Atos / Eviden	Global systems integrator	sells Machine learning	structural	—
NiCE	Customer-experience platform	sells Machine learning — also an Orange partner	structural	—
Databricks	Data and AI platform	sells Machine learning	structural	—
Dataiku	Data and AI platform	sells Machine learning	structural	—
Palantir	Data and AI platform	sells Machine learning; competes in OX: Smart Industries	structural	—

Schneider Electric	Industrial / OT specialist	active in Construction; competes in OX: Smart Industries	structural	—
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evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 Have you experienced any air quality-related health incidents or near-misses on your construction sites in the past year?
- 2 Do you currently monitor air quality on any of your sites, and if so, what parameters and how frequently?
- 3 What is your process for deciding when to halt work due to poor air quality, and who is responsible?
- 4 Are you subject to any specific regulations or client requirements regarding worker exposure to airborne contaminants?
- 5 How would you prioritize a solution that predicts health risks versus one that only alerts on current conditions?

Objections you will meet

They will say	You can answer
We already have basic air quality monitors on site.	That's a good starting point. Our solution builds on that by using machine learning to predict risks before they become dangerous, not just react to current readings.
This sounds expensive and complex.	We understand cost is a concern. We can start with a pilot on one site to demonstrate value, and our IoT platform is designed to integrate with existing sensors, minimizing upfront investment.
Why not just use AWS or Google Cloud for this?	They offer strong ML tools, but we provide the full package: reliable connectivity via private mobile networks, IoT integration, and local expertise. We can also work with those platforms if you prefer.

Risks and unknowns

The main risk is that construction firms may not yet see air quality monitoring as a priority, or may prefer to work with a hyperscaler directly. Unknowns include the accuracy of ML models in diverse site conditions and the willingness of customers to invest in new safety technology. The evidence base is academic; real-world deployments are not yet documented.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a construction safety monitoring solution, integrating Live Objects and private mobile networks, and referencing AWS, Google Cloud, NVIDIA, and NiCE as technology partners.
Sales	In a meeting with a construction firm's safety director, say: 'Orange can help you protect your workers with real-time air quality monitoring and predictive risk alerts, using our Live Objects IoT platform and private mobile networks for reliable connectivity.'
Strategist / Innovator	Commission a deep-dive brief on ML-based air quality monitoring for construction worker safety, leveraging Orange Group researchers and IoT experts to assess feasibility and IP positioning.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-1C4F366EE4AA	2026-07-09	ted.europa.eu	1	Germany – Environmental indicators analysis other than for construction – One Health in der Pra...
SIG-AC86A5D91D50	2026-06-11	ted.europa.eu	1	France – Pollution tracking and monitoring and rehabilitation – SPCH a Harbonnieres (80) - Etud...
SIG-1CE665656A70	2026-05-20	ted.europa.eu	1	Finland – Repair and maintenance services of building installations – Ilmanvaihtojarjestelmien ...
SIG-36DA075F2A6A	2026-03-02	openalex.org	1	IoT-based Industrial Safety and Monitoring System with Real-Time Alert Mechanism
SIG-9528B2CFF302	2026-01-01	openalex.org	1	Exploring Organizational and Individual Determinants of Construction Workers' Safety Behavior: ...
SIG-A33E4A343E60	2025-12-17	openalex.org	1	Immersive technologies for evaluating industrial safety training in high-risk environments: a r...
SIG-6ADB457BBC36	2025-11-19	openalex.org	1	A Machine Learning Approach to Predicting Air Quality and Health Risks for Construction Worker ...
SIG-3E11F4C71251	2025-11-13	openalex.org	1	Artificial Intelligence (AI) in Construction Safety: A Systematic Literature Review
SIG-8BC657BF5DA5	2025-10-04	openalex.org	1	Assessing workers' neuro-physiological stress responses to augmented reality safety warnings in...
SIG-80CAA03177B8	2025-09-26	openalex.org	1	Occupational health risks, safety essentials, and safety beliefs among construction workers in ...
SIG-C3E338C2C7D2	2025-09-15	openalex.org	1	Using Safety-Specific Transformational Leadership to Improve Safety Behavior Among Construction...
SIG-72037039EB81	2025-09-12	openalex.org	1	Safety Violations of Construction Workers: An Empirical Study

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:16+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:07+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.